

Mini-Belt Characterization System using Impedance Spectroscopy

Project type: ☐ Hardware ☐ Software ☒ Hardware/Software ☐ Simulation ☒ AI/ML

Project Overview

The Mini-Belt Characterization System is an automated, in-line material analysis platform that combines STM32-based Electrochemical Impedance Spectroscopy (EIS) with a miniature conveyor belt and Interdigitated Electrodes (IDEs). The system performs real-time quality monitoring of moving materials (e.g., moisture content, freshness, curing state) without stopping the production line.

Instead of static lab measurements, this system utilizes Multisine broadband excitation to capture high-speed impedance fingerprints as samples pass over the characterization zone. The architecture is designed for research into smart manufacturing and food processing applications.

Project Tasks

3.1 Miniature Conveyor Hardware

- Compact belt assembly driven by stepper motor for precise speed control
- "Floating header" sensor mount for consistent IDE contact with moving samples
- Optical encoder for position tracking and timestamp correlation

3.2 Multisine Signal Generation

- Optimized broadband excitation (1 Hz to 10 kHz) using STM32 DAC
- DMA-driven waveform output for jitter-free synthesis
- Crest factor minimization via phase optimization algorithms

3.3 EIS Acquisition & Processing

- High-speed ADC sampling of voltage and current with precise phase alignment
- On-chip FFT-based DFT computation to extract complex impedance in real time
- Nyquist and Bode plot generation on Python host

3.4 Belt Speed & Throughput Sync

- Correlate measurement timestamps with belt position using optical encoder
- Dynamically adjust sampling rates based on conveyor speed
- Impedance vs. belt position heatmaps for spatial mapping

3.5 Host Interface & Machine learning & Visualization

- Python-based GUI (matplotlib/pyqtgraph) for real-time Nyquist/Bode plots
- Real-time characterization using Machine learning
- Data streaming via UART/USB-CDC

Expected Outcomes

-  A stable and fully functional system

Competences Developed

- Embedded systems programming (C/C++ on STM32F4/F7)
- Analog signal conditioning for bioimpedance measurements
- Synchronous demodulation and lock-in amplifier algorithms
- FFT/Goertzel algorithm implementation in firmware
- Python desktop application development with spectral analysis

Notice: The use of ChatGPT and other LLM services susceptible to data breaching is strictly forbidden*

* Except: Academic Cloud, Local LLM, and Telemetry-disabled Claude.

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